8.3 Odziv diskretnog sustava pomoću matrica

$x[0] \neq 0$

$x[n+1] = A \cdot x[n] + B\,u[n]$

$y[n] = C\,x[n] + D\,u[n]$

$x[1] = A\,x[0] + B\,u[n]$

$x[2] = A\,x[1] + B\,u[1] = A\left(A\,x[0] + B\,u[0]\right) + B\,u[1] = A^2 x[0] + AB\,u[0] + B\,u[1]$

$x[3] = A\,x[2] + B\,u[2] = A\left(A^2 x[0] + A\,B\,u[0] + B\,u[1]\right) + B\,u[2]$

$= A^3 x[0] + A^2 B\,u[0] + A\,B\,u[1] + B\,u[2]$

$\vdots$

$$x[n] = A^n x[0] + \sum_{m=0}^{n-1} A^{(n-1-m)} B\,u[m]$$

$$x[n] = \begin{cases} x[0] & , \quad n = 0 \\ \underbrace{A^n x[0]}_{\text{repobuđeni}} + \underbrace{\sum_{m=0}^{n-1} A^{n-m-1} B\,u[m]}_{\text{odziv vrijeme}} & , \quad n \geq 1 \end{cases}$$

odziv stanja sustava

$\Phi[n] = A^n$ prijelazna matrica / fundamentalna

slično kod KONT (tamo je Φ)

$$x[n] = \begin{cases} x[0] & n = 0 \\ \Phi[n] \cdot x[0] + \sum_{m=0}^{n-1} \Phi[n-m-1] \cdot B \cdot u[m] & , \quad n \geq 1 \end{cases}$$

$$y[n] = C\,x[n] + D\,u[n]$$

$$= \begin{cases} C\,x[0] + D\,u[0] & n = 0 \\ C A^n x[0] + C \sum_{m=0}^{n-1} A^{n-m-1} B\,u[m] + D\,u[n] & , \quad n \geq 1 \end{cases}$$

$$y[n] = \begin{cases} C\,x[0] + D\,u[0] & , \quad n = 0 \\ C\,\Phi[n] \cdot x[0] + \sum_{m=0}^{n-1} C\,\Phi[n-m-1] \cdot B \cdot u[m] + D\,u[n] & , \quad n \geq 1 \end{cases}$$

Češći to koristimo u Z-domeni

$$x[n+1] = Ax[n] + Bu[n]$$

ne zaboravi da su matrice

$$z\,X(z) - z\,x[0] = A\,x[z] + B\,u[z]$$

z je broj! → napisati u obliku mat

↳ $z \cdot I$

$$(zI - A)\,x[z] = B\,u(z) + z\,x[0] \quad / (zI-A)^{-1} \text{ (ulijevo)}$$

budući da radimo s matricama bitno je s koje strane dijelimo!

$$X[z] = (zI-A)^{-1} \cdot B \cdot u[z] + (zI-A)^{-1} \cdot \underline{z} \cdot x[0]$$

skalar, može naprijed

$$X[z] = \underbrace{z(zI-A)^{-1}}_{\Phi(z)\ (\text{od } \Phi[n] = A^n)} \cdot x[0] + \underbrace{(zI-A)^{-1} \cdot B \cdot u[z]}_{\text{odziv stanja mirnog sustava}}$$

↳ z transf. fundamentalne (prijelazne) mat

$$\underbrace{\Phi_z = z(zI-A)^{-1}}_{\text{odziv stanja nepobuđenog sust}}$$

$$y[z] = Cx[z] + Du[z]$$

$$= C \cdot z(zI-A)^{-1} \cdot x[0] + C \cdot (zI-A)^{-1} \cdot B \cdot u[z] + Du[z]$$

$$= \underbrace{C \cdot z(zI-A)^{-1} \cdot x[0]}_{\text{Odziv nepobuđ. sustava}} + \underbrace{\left[C \cdot (zI-A)^{-1} B + D\right]}_{\text{PRIJENOSNA MAT}} u[z]$$

↳ odziv mirnog sustava

$$H(z) = C(zI-A)^{-1}B + D$$

npr.

$$\begin{array}{cc} \text{1. ulaz} & \text{2. ulaz} \end{array}$$

$$\begin{bmatrix} \frac{1}{z+1} & \frac{2}{z+3} \\ \frac{4}{z+1} & \frac{5}{z+1} \end{bmatrix} \begin{array}{l} \leftarrow \text{1 izlaz} \\ \leftarrow \text{2. izlaz.} \end{array}$$

→ 2 ulaza i 2 izlaza

impulsni odziv $h[n]$ → [ ]

* inverz:

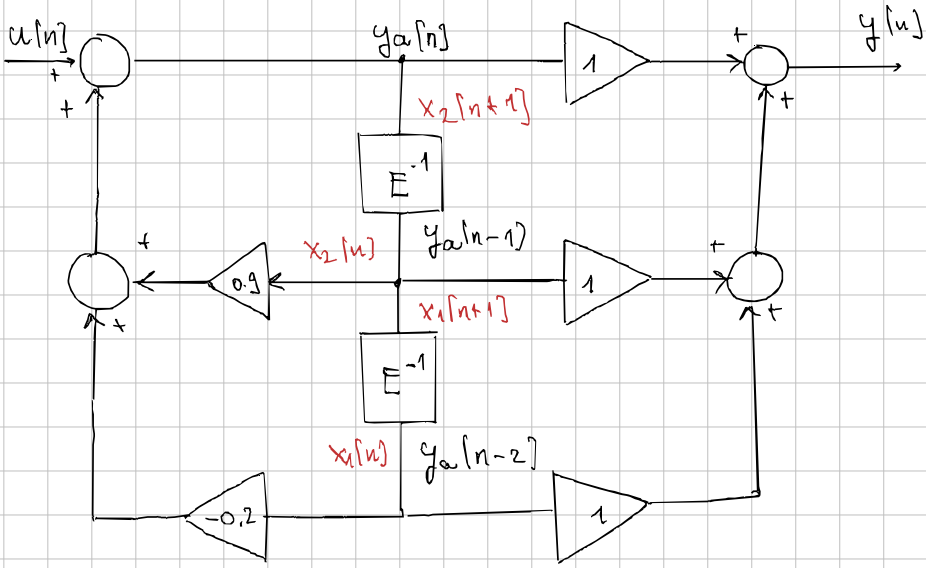
$$(zI-A)^{-1} = \frac{\operatorname{adj}(zI-A)}{\det(zI-A)}$$

$\det(zI-A) = 0$ daje polove

↳ karakt. jed. sustava

Primjer: $y[n] \overset{a_1}{-0.9}\, y[n-1] \overset{a_2}{+0.2}\, y[n-2] = u[n] + u[n-1] + u[n-2]$

→ za dobit var stanja najlakše nacrtati direktnu II. realizaciju



$$x_1[n+1] = x_2[n]$$

$$x_2[n+1] = y_a[n] \overset{\text{na ulazu samo } u[n]}{=\!=} 0.9\underbrace{y_a[n-1]}_{x_2[n]} - 0.2\underbrace{y[n-2]}_{x_1[n]} + u[n]$$

$$= \underline{0.9\, x_2[n] - 0.2\, x_1[n] + u[n]}$$

$$y[n] = y_a[n] + y_a[n-1] + y_a[n-2] = x_1[n] + x_2[n] + y_a[n]$$

$$= x_1[n] + x_2[n] + 0.9\, x_2[n] - 0.2\, x_1[n] + u[n]$$

$$= 0.8\, x_1[n] + 1.9\, x_2[n] + u[n]$$

$$\boxed{\begin{aligned} x[n+1] &= \begin{bmatrix} x_1[n+1] \\ x_2[n+1] \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -0.2 & 0.9 \end{bmatrix}}_{A} \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u[n] \\ y[n] &= \underbrace{\begin{bmatrix} 0.8 & 1.9 \end{bmatrix}}_{C} \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix} + \underbrace{1}_{D} \cdot u[n] \end{aligned}}$$

$$x[n+1] = \begin{bmatrix} x_1[n+1] \\ x_2[n+1] \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -0.2 & 0.9 \end{bmatrix}}_{A} \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u[n]$$

$$y[n] = \underbrace{\begin{bmatrix} 0.8 & 1.9 \end{bmatrix}}_{C} \begin{bmatrix} x_1[n] \\ x_2[n] \end{bmatrix} + \underbrace{1}_{D} \cdot u[n]$$

↳ pomoću toga nalazimo odziv sustava

→ trebamo poč. vr => $x[0]$ ali zadano!

$$x[0] = \begin{bmatrix} 2.593 \\ 2.0659 \end{bmatrix}$$

i $u[n]$ mora biti zadano → $u[n] = -9 \cdot (-0.9)^n \mu[n]$

$$U(z) = -9 \frac{z}{z+0.9}$$

* pomoću

$y[-1] = 6$ $\quad u[-1] = 0$

$y[-2] = -3$ $\quad u[-2] = 0$

možemo dobiti $x[0]$

↳ iz poznatih vrijednosti dobivaju se poč. stanja

* sjeti se: $Y(z) = Cz(zI-A)^{-1} x[0] + [C(zI-A)^{-1} B + D] U(z)$

$Y_0(z) = Cz(zI-A)^{-1} x[0]$ → prvo napravimo

$\Phi(z) = z(zI-A)^{-1}$ → fundamentalna matrica

$$\Phi(z) = z\left(\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -0.2 & 0.9 \end{bmatrix}\right)^{-1} = z \begin{bmatrix} z & -1 \\ 0.2 & z-0.9 \end{bmatrix}^{-1} = z \cdot \frac{1}{z(z-0.9)+0.2} \begin{bmatrix} z-0.9 & 1 \\ -0.2 & z \end{bmatrix}$$

$$\Phi(z) = \frac{z}{z^2-0.9z+0.2} \begin{bmatrix} z-0.9 & 1 \\ -0.2 & z \end{bmatrix}$$

$$\hookrightarrow Y_0(z) = \begin{bmatrix} 0.8 & 1.9 \end{bmatrix} \frac{z}{z^2-0.9z+0.2} \begin{bmatrix} z-0.9 & 1 \\ -0.2 & z \end{bmatrix} \begin{bmatrix} 2.593 \\ 2.0659 \end{bmatrix}$$

$$Y_0(z) = \frac{z}{z^2-0.9z+0.2} \begin{bmatrix} 0.8(z-0.9) - 1.9 \cdot 0.2 & 0.8 \cdot 1 + 1.9z \end{bmatrix} \begin{bmatrix} 2.593 \\ 2.0659 \end{bmatrix}$$

$$= \frac{z}{z^2-0.9z+0.2} \left[(0.8z - 1.1) \cdot 2.593 + (0.8 + 1.9z) \cdot 2.0659 \right]$$

$$Y_0(z) = \frac{6z^2 - 1.2z}{(z-0.4)(z-0.5)} \Longrightarrow \boxed{y_0[n] = \left(18(0.5)^n - 12(0.4)^n\right) \mu[n]}$$

- Sad tražimo mirni: $y_m[n] = \left[C(zI-A)^{-1}B + D \right] u[z]$

$$y_m[z] = \left[C\,\overset{z^{-1}}{\phi[z]} \cdot B + D \right] u[z]$$

$$= \left(\frac{\cancel{z}}{z^2-0.9z+0.2} \begin{bmatrix} 0.8z-1.1 & 0.8+1.9z \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \right) \left(-9 \cdot \frac{z}{z+0.9} \right)$$

$$y_m[z] = \left(\frac{\cancel{z}\,1}{z^2-0.9z+0.2} \cdot (0.8+1.9z) + 1 \right) \cdot (-9) \frac{z}{z+0.9}$$

$$y_m[z] = \dots = \frac{z^2+z+1}{z^2-0.9z+0.2} \cdot \frac{-9z}{z+0.9} = \frac{-112.5z}{z-0.5} + \frac{108z}{z-0.4} - \frac{4.5z}{z+0.9}$$

$$y_m[n] = \left(-112.5\,(0{,}5)^n + 108\,(0{,}4)^n - 4.5\,(-0.9)^n \right) \mu[n]$$

$$\boxed{y[n] = y_0[n] + y_m[n] = \left(-94.5\,(0{,}5)^n + 96\,(0.4)^n - 4.5\,(-0.9)^n \right) \mu[n]}$$